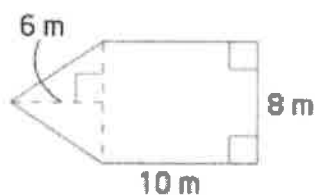


Area of Composite Figures Worksheet

MPM1D

1. Find the area of each of the composite figures:

a)



$$A_{\text{triangle}} = \frac{1}{2}bh$$

$$= \frac{1}{2}(8)(6)$$

$$= 24 \text{ m}^2$$

$$A_{\text{rectangle}} = (l)(w)$$

$$= (10)(8)$$

$$= 80 \text{ m}^2$$

$$\text{Total Area} = 24 + 80$$

$$= 104 \text{ m}^2$$

b)



height of triangle:

$$a^2 = c^2 - b^2$$

$$a^2 = 13^2 - 12^2$$

$$a^2 = 25$$

$$a = 5$$

$$A_{\text{triangle}} = \frac{1}{2}(12)(5)$$

$$= 30 \text{ m}^2$$

$$A_{\text{square}} = l^2$$

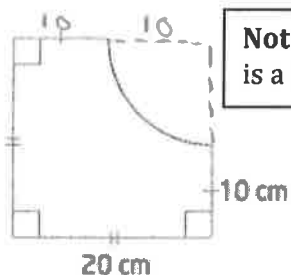
$$= (12)^2$$

$$= 144 \text{ m}^2$$

$$\text{Total Area} = 30 + 144$$

$$= 174 \text{ m}^2$$

c)



Note: the cut-out is a quarter circle

$$A_{\text{square}} = l^2$$

$$= 20^2$$

$$= 400 \text{ cm}^2$$

$$A_{\frac{1}{4}\text{circle}} = \frac{1}{4}\pi r^2$$

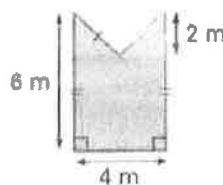
$$= \frac{1}{4}\pi(10)^2$$

$$= 78.5 \text{ cm}^2$$

$$\text{Total area} = 400 - 78.5$$

$$= 321.5 \text{ cm}^2$$

d)



$$A_{\text{rectangle}} = (l)(w)$$

$$= (4)(6)$$

$$= 24 \text{ m}^2$$

$$A_{\text{triangle}} = \frac{1}{2}bh$$

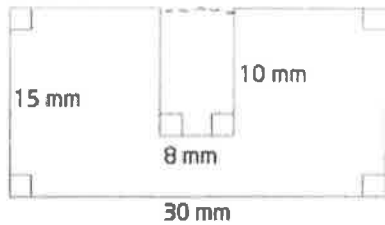
$$= \frac{1}{2}(4)(2)$$

$$= 4 \text{ m}^2$$

$$\text{Total area} = 24 - 4$$

$$= 20 \text{ m}^2$$

e)

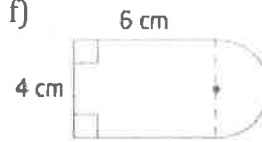


$$\begin{aligned} A_{\text{large rectangle}} &= (l)(w) \\ &= (30)(15) \\ &= 450 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} A_{\text{cut out rectangle}} &= (l)(w) \\ &= (8)(10) \\ &= 80 \text{ mm}^2 \end{aligned}$$

$$\begin{aligned} \text{Total Area} &= 450 - 80 \\ &= 370 \text{ mm}^2 \end{aligned}$$

f)



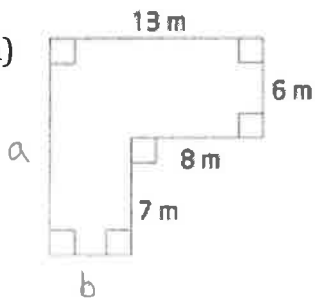
$$\begin{aligned} A_{\text{rectangle}} &= (l)(w) \\ &= (4)(6) \\ &= 24 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} A_{\text{semicircle}} &= \frac{1}{2} \pi r^2 \\ &= \frac{1}{2} \pi (2)^2 \\ &= 6.3 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Total area} &= 24 + 6.3 \\ &= 30.3 \text{ cm}^2 \end{aligned}$$

2. Find the perimeter of each of the composite figures:

a)

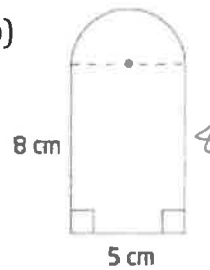


$$\begin{aligned} a &= 7 + 6 \\ &= 13 \text{ m} \end{aligned}$$

$$\begin{aligned} b &= 13 - 8 \\ &= 5 \text{ m} \end{aligned}$$

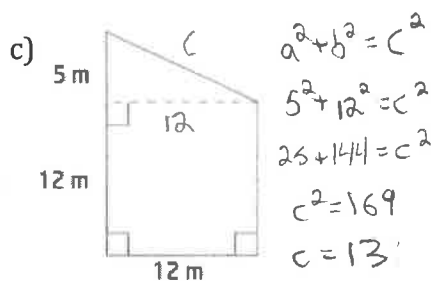
$$\begin{aligned} \text{Perimeter} &= 13 + 6 + 8 + 7 + 5 + 13 \\ &= 52 \text{ m} \end{aligned}$$

b)



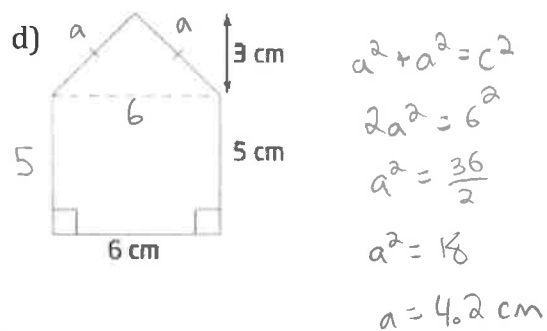
$$\begin{aligned} \text{Perimeter of semicircle} &= \frac{1}{2} \pi d \\ &= \frac{1}{2} \pi (5) \\ &= 7.9 \text{ cm} \end{aligned}$$

$$\begin{aligned} \text{Perimeter} &= 8 + 5 + 8 + 7.9 \\ &= 28.9 \text{ cm} \end{aligned}$$



$$\text{Total Perimeter} = 12 + 12 + 12 + 5 + 13$$

$$= 54 \text{ m}$$



$$\text{Total Perimeter} = 5 + 6 + 5 + 4.2 + 4.2$$

$$= 24.4 \text{ cm}$$

3. a) What length of fencing is needed to surround this yard?

$$c^2 = a^2 + b^2$$

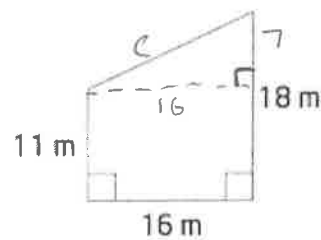
$$c^2 = 16^2 + 7^2$$

$$c^2 = 305$$

$$c = 17.5$$

$$\text{Total Perimeter} = 11 + 16 + 18 + 17.5$$

$$= 62.5 \text{ m}$$



b) What is the area of the yard?

$$A_{\text{rectangle}} = (l)(w)$$

$$= (11)(16)$$

$$= 176 \text{ m}^2$$

$$A_{\text{triangle}} = \frac{1}{2}bh$$

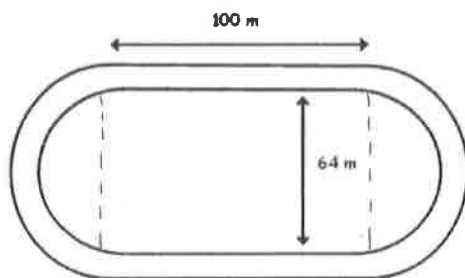
$$= \frac{1}{2}(16)(7)$$

$$= 56 \text{ m}^2$$

$$\text{Total Area} = 176 + 56$$

$$= 232 \text{ m}^2$$

4. Here is a diagram of the track King's is thinking of adding around the new field. It consists of two parallel lines and a semicircle at each end. The track is 10 meters wide.



- a) If someone runs one lap on the inside of the track, how far will they have run?

$$\begin{aligned}\text{Circumference of circle} &= \pi d \\ &= \pi (64) \\ &= 201.1 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Perimeter of track} &= 100 + 100 + 201.1 \\ &= 401.1 \text{ m.}\end{aligned}$$

- b) If someone runs one lap on the outside of the track, how far will they have run?

$$\begin{aligned}\text{Circumference of circle} &= \pi d \\ &= \pi (84) \\ &= 263.9 \text{ m}\end{aligned}$$

$$\begin{aligned}\text{Perimeter} &= 100 + 100 + 263.9 \\ &= 463.9 \text{ m}\end{aligned}$$

- c) Find the difference between the distances of running on the inside or outside of the track.

$$463.9 - 401.1 = 62.8 \text{ m.}$$